Set 1: Mirror reflections: what a reflection must show

Local outcome LOGIC.SPATIAL.01: Decide whether an image is consistent with a reflection in a horizontal or vertical mirror line

Puzzle set · genre: reflection and symmetry · MP7, MP6

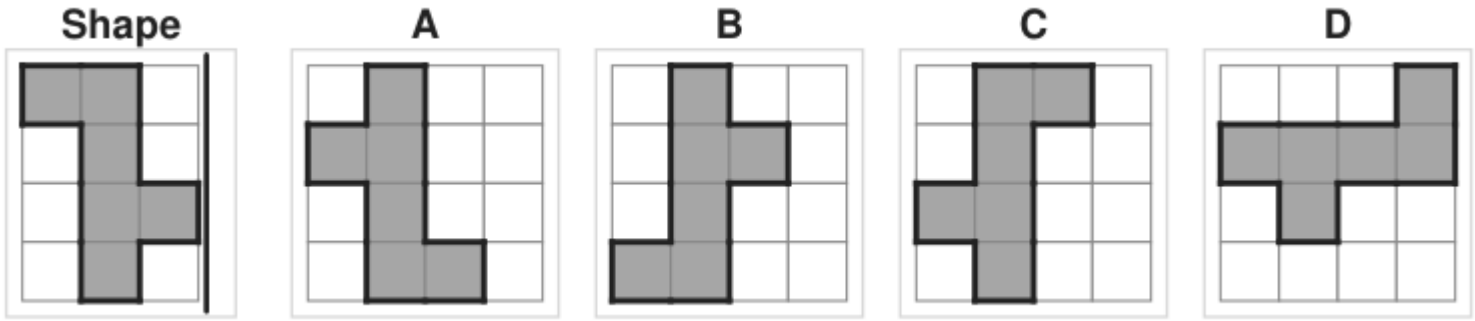
MAIN: Decide what a mirror reflection must show

BACKWARD 1: Strategy: reflect one cell at a time

FORWARD 1: ADVANCED APPLICATION: Combine reflections and read mirrored clocks and words

FORWARD 2: EXPERT CHALLENGE: Construct and count symmetric figures

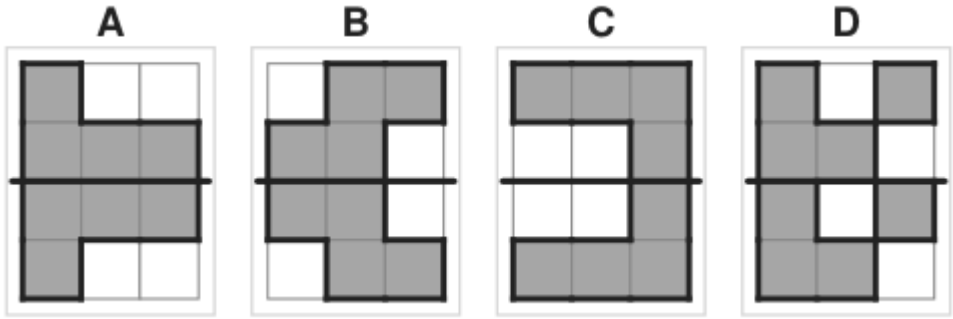
The shaded shape is reflected in the dashed mirror line on its right. Which figure shows its mirror image? (Only the shape matters, not where it sits in its box.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Each figure shows a shape above a dashed water line and, below the line, what is supposed to be its reflection in the water. In which figure can the lower shape NOT be the reflection of the upper one?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Words are written left to right in block capitals and seen in a left-right mirror (a vertical mirror line). In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same in the mirror?

A. TOOT
B. ZOOM
C. NOON
D. BIB

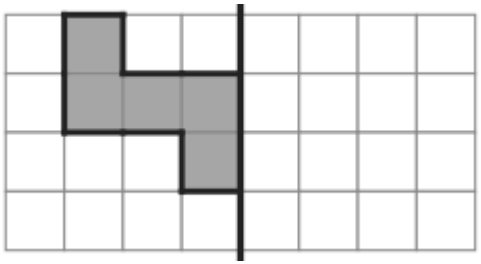
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A word is written in block capitals and seen reflected in calm water below it (a horizontal mirror line). In the block capitals used here, exactly these letters look the same in a top-bottom mirror: B C D E H I K O X. The letter order stays the same in this reflection. Which word looks exactly the same in the water?

A. BEAD
B. CODE
C. CORE
D. BOXES

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The grid has 4 rows and 8 columns. The dashed mirror line runs between columns 4 and 5. The shaded cells are in row 1 column 2, row 2 columns 2, 3 and 4, and row 3 column 4. Mark the cells of their mirror image. (Cells are named row,column; row 1 is the top row and column 1 the left column.)



Mark cells on the 4 × 8 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A grid has 10 columns. A vertical mirror line lies between columns 5 and 6. A shaded cell is in column 2. In which column is its mirror image?

Column: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A grid has 8 rows. A horizontal mirror line lies between rows 4 and 5. A shaded cell is in row 1. In which row is its mirror image?

Row: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A reflection in a vertical mirror line keeps every cell in its own row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A cell is 3 columns to the left of a vertical mirror line. Where is its mirror image?

A. 6 columns to the right of the line
B. on the mirror line itself
C. 3 columns to the left of the line
D. 3 columns to the right of the line

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A row of 6 cells has a mirror line between cells 3 and 4. Cell 1 is shaded. Mark the cell of its mirror image. (The grid is 1 row by 6 columns; cells are named row,column.)

Mark cells on the 1 × 6 grid (on screen: click each cell; cells are named row,column).

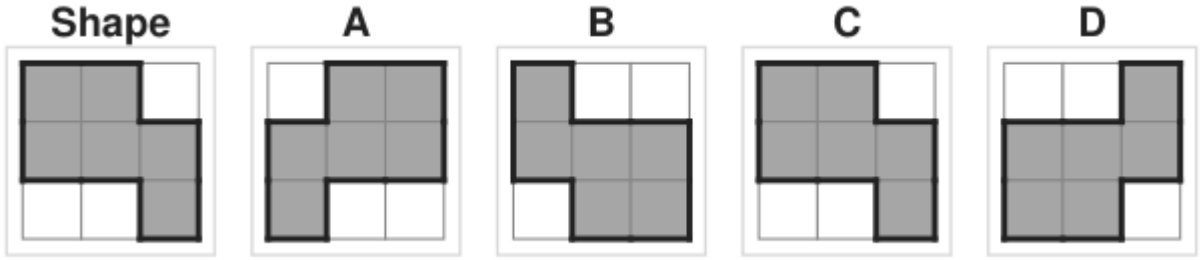
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A clock face with no numbers shows 2:20. You see it in a left-right mirror. What time does the mirrored clock appear to show?

A. 10:20
B. 2:40
C. 9:40
D. 3:40

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The shape is reflected in a vertical mirror line and then the result is reflected in a horizontal mirror line. Which figure shows the final result? (Only the shape matters, not where it sits.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

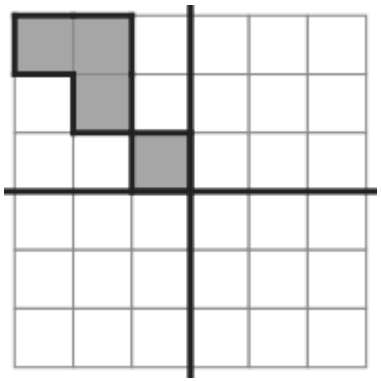
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A word is written DOWN the page, one block capital under another, and seen in a left-right mirror. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same?

A. MATH
B. MAZE
C. BATH
D. MASK

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A 6 by 6 grid has two dashed mirror lines: a vertical one between columns 3 and 4 and a horizontal one between rows 3 and 4. The figure must look the same in BOTH mirrors. Four cells are already shaded in the top-left quarter: (1,1), (1,2), (2,2) and (3,3). Mark every shaded cell of the completed figure, including the four given cells.



Mark cells on the 6 × 6 grid (on screen: click each cell; cells are named row,column).

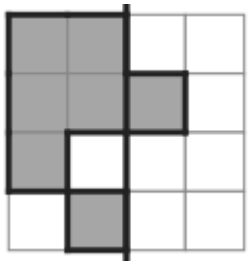
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A figure looks the same after a reflection in a vertical line through its center and after a reflection in a horizontal line through its center. Then it also looks the same after a half-turn about its center.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The figure is on a 4 by 4 grid with a dashed vertical line between columns 2 and 3. Shaded cells can be added but not removed. What is the least number of cells you must add so that the figure looks the same in a mirror along the dashed line?



Cells to add: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Strings of 3 letters are made from the letters A, H, M, O and T (letters may repeat). Each is written left to right in block capitals. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. How many of these strings look exactly the same in a left-right mirror?

Number of strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Two vertical mirror lines are 3 cells apart: line 1 on the left, line 2 on the right. A shape is reflected in line 1 and the image is then reflected in line 2. Where does the final image end up?

A. a mirror image of the shape, 6 cells to the right
B. back exactly where the shape started
C. the original shape moved 3 cells to the right, not flipped
D. the original shape moved 6 cells to the right, not flipped

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

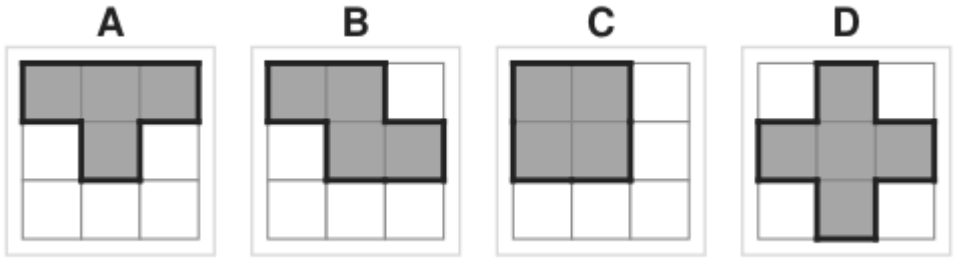
A row of 12 cells is numbered 1 to 12. A mirror line lies between cells 6 and 7. Give two cell numbers a and b, with a less than b, that are mirror images of each other.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Count only mirror lines that run vertically, horizontally or diagonally through the center of the shape (the middle of the smallest rectangle around it). Which figure has exactly ONE mirror line?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Set 2: Read exactly what is stated

Local outcome LOGIC.READING.01: Read negations, double negations, units and relationships exactly as stated

Puzzle set · genre: precise reading · MP6, MP1

MAIN: Read negations, units and relationships exactly as stated
BACKWARD 1: Strategy: rewrite each statement in plain form
FORWARD 1: ADVANCED APPLICATION: Read layered negations, units and family clues
FORWARD 2: EXPERT CHALLENGE: Precise reading in multi-step puzzles

Three students each make one statement about themself, and every statement is true. Dee: "It is not true that I finished the race." Eli: "I did not fail to finish the race." Fay: "I never said that I finished the race." Who definitely finished the race?

A. Eli and Fay
B. Eli only
C. Dee and Fay
D. Dee only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q1
Logic, Set 2: Read exactly what is stated · MAIN · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Shoelaces are sold only in packs of 2 laces for $3 a pack. Each sneaker needs exactly one lace. A club needs laces for 12 sneakers and also buys 4 whistles at $5 each. What is the least the club can spend, in dollars?

Dollars: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q2
Logic, Set 2: Read exactly what is stated · MAIN · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

In the Ortiz family, each son has exactly one brother, and each daughter has exactly two brothers and three sisters. How many children are in the family?

Children: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q3
Logic, Set 2: Read exactly what is stated · MAIN · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A box holds only red pens and blue pens: at least 3 red pens, at most 2 blue pens, and exactly 6 pens in all. Which statement MUST be true?

A. There are exactly 2 blue pens.
B. There are exactly 3 red pens.
C. There are at least 4 red pens.
D. There are no blue pens.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q4
Logic, Set 2: Read exactly what is stated · MAIN · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

On this team, April, June and May are the names of three players (not months). A coach writes: "June scored after April and before May." Put the players in the order in which they scored, first to last.

Put these in order:

A April
J June
M May

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q5
Logic, Set 2: Read exactly what is stated · MAIN · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Treat "unhappy" as meaning exactly "not happy". What does "Rui is not unhappy" say?

A. Rui is unhappy.
B. Nothing at all about Rui.
C. Rui is sad and tired.
D. Rui is happy.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q1
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A drawer holds 18 single socks that make complete matching pairs. How many pairs is that?

Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q2
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"At most 3 people may enter" allows exactly 3 people to enter.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q3
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"Not all of the doors are locked." Which statement means the same?

A. At least one door is unlocked.
B. No door is locked.
C. Every door is unlocked.
D. Exactly one door is unlocked.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q4
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Lin has 3 brothers. Each of her brothers has exactly one sister. How many children are in the family?

Children: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q5
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Every statement below is true. Which person is going to the game? Gus: "It is not true that I am not going." Hal: "I am not not staying home." Ivy: "It is false that I won't go."

A. Hal and Ivy
B. Gus and Ivy
C. Gus only
D. all three

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q1
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Eggs are sold in half-dozen boxes (6 eggs) for $2 a box. A recipe uses 4 eggs per cake, and a baker makes 9 cakes. What is the least she can spend on eggs, in dollars?

Dollars: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q2
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

In the Novak family, each son has exactly 3 sisters and 2 brothers. Each daughter has exactly 2 sisters. How many children are in the family?

Children: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q3
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A bag holds only red and blue marbles: at least twice as many red as blue, at least 3 blue, and at most 10 marbles in all. Which statement MUST be true?

A. There are exactly 10 marbles.
B. There are exactly 6 red marbles.
C. There are exactly 3 blue marbles.
D. There are more blue marbles than red ones.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q4
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"Only members may vote" means that everyone who votes is a member.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q5
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Four statements: (1) "It is not the case that 7 is not odd." (2) "It is false that 10 is not even." (3) "It is not true that 9 is not a multiple of 4." (4) "5 is not unequal to 5." How many of the four statements are true?

Number true: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q1
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

On a quiz show, Monday, Tuesday, Wednesday and Friday are the first names of four contestants. The host says: "Wednesday answered before Monday. Friday answered right after Tuesday. Monday did not answer last." Put the contestants in answering order, first to last.

Put these in order:

Mo Monday
Tu Tuesday
We Wednesday
Fr Friday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q2
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A bag holds only red and blue marbles: the number of red marbles is at least twice the number of blue, there are at least 3 blue marbles, and there are at most 10 marbles in all. Give a possible number of red marbles r and blue marbles b.

red r = _____

blue b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q3
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

In a group of brothers and sisters, every girl has exactly as many brothers as sisters, and there are 3 boys. How many girls are there?

Girls: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q4
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"No student who did not study passed the test." Which statement follows?

A. Every student who passed studied.
B. Some students who studied failed.
C. Every student who studied passed.
D. No student passed.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q5
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Set 3: Statements, premises and conclusions

Local outcome LOGIC.ARGUMENT.01: Identify statements, premises, conclusions and unstated assumptions in an argument

Formal logic set · genre: formal logic · MP3

- MAIN: Identify statements, premises, conclusions and assumptions
- BACKWARD 1: Which sentences are statements?
- BACKWARD 2: Find the conclusion by its signal words
- FORWARD 1: ADVANCED APPLICATION: Arguments with several premises and hidden structure
- FORWARD 2: EXPERT CHALLENGE: Evaluate structure and missing links

"The trail is closed whenever it rains. It is raining now. So the trail is closed." Which sentence is the conclusion?

A. The trail is closed whenever it rains.
B. Rain is likely tomorrow.
C. It is raining now.
D. The trail is closed.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q1
Logic, Set 3: Statements, premises and conclusions · MAIN · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which of these is a statement (a sentence that is either true or false)?

A. Please close the window.
B. Every square has four sides.
C. Is 15 a prime number?
D. What a great game!

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q2
Logic, Set 3: Statements, premises and conclusions · MAIN · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Since every ticket holder may enter, and Zoe holds a ticket, Zoe may enter." Which sentence is a premise?

A. Only Zoe may enter.
B. Zoe may enter.
C. Zoe holds a ticket.
D. Zoe bought the ticket online.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q3
Logic, Set 3: Statements, premises and conclusions · MAIN · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Ravi will be on time, because the bus left on schedule." Which unstated assumption does this argument need?

A. The bus is red.
B. Ravi was late yesterday.
C. Some buses are late.
D. Ravi is on that bus.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q4
Logic, Set 3: Statements, premises and conclusions · MAIN · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"All of our club's robots passed inspection. Unit 7 is one of our club's robots. The room is cold. Therefore Unit 7 passed inspection." Which sentence plays no part in the reasoning?

A. The room is cold.
B. Unit 7 is one of our club's robots.
C. All of our club's robots passed inspection.
D. Therefore Unit 7 passed inspection.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q5
Logic, Set 3: Statements, premises and conclusions · MAIN · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Close the door." is a statement.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q1
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"7 is greater than 9." is a statement.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q2
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which is NOT a statement?

A. Nobody scored.
B. Who won the game?
C. Our team won.
D. The game ended in a tie.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q3
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"x + 2 = 5" with no value of x given is a statement.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q4
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which sentence is a statement?

A. Visit Paris.
B. Is Paris far?
C. Paris is in Spain.
D. Let's go to Paris!

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q5
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which word usually introduces a conclusion?

A. given that
B. therefore
C. because
D. since

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q1
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which word usually introduces a reason (a premise)?

A. thus
B. consequently
C. because
D. so

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q2
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"The lake is frozen, so we can skate." What is the conclusion?

A. Skating is fun.
B. The lake is frozen.
C. It is winter.
D. We can skate.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q3
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"We should leave now, since the show starts at 7." What is the conclusion?

A. We should leave now.
B. We are late.
C. The show is long.
D. The show starts at 7.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q4
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

In "Because the battery is dead, the car will not start," the conclusion is "the battery is dead."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q5
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"The museum must be open today. After all, it is open every weekday except Monday, and today is Thursday." Which sentence is the conclusion?

A. The museum must be open today.
B. It is open every weekday except Monday.
C. The museum is closed on Monday.
D. Today is Thursday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q1
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Put these sentences in the order of a clear argument: the general rule first, then the fact, then the conclusion.

Put these in order:

A Therefore, 91 is not prime.

B Any whole number greater than 1 with a factor other than 1 and itself is not prime.

C 91 = 7 x 13, so 91 has the factor 7.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q2
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Coach Lee will pick Maya, because Coach Lee always picks the fastest runner." Which assumption is needed?

A. Maya ran yesterday.
B. Coach Lee likes Maya.
C. Maya wants to be picked.
D. Maya is the fastest runner.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q3
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

An argument can have a true conclusion even when one of its premises is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q4
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"If the alarm is set, it rings at 6. It did not ring at 6. Hence it was not set." How many premises does this argument have?

A. 0
B. 2
C. 3
D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q5
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"All sprinters on the team train daily. So Kim trains daily." Which added premise makes the conclusion follow for certain?

A. Kim trains often.
B. Kim is on the team.
C. Kim is a sprinter on the team.
D. Some sprinters train daily.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q1
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every number divisible by 6 is even. So 84 is even." Which added premise makes this valid?

A. 84 is greater than 6.
B. 84 is divisible by 4.
C. Some even numbers are divisible by 6.
D. 84 is divisible by 6.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q2
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

In a valid argument, adding an extra true premise can make the conclusion no longer follow.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q3
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Arrange these sentences into an argument that ends with its conclusion: chain the two rules.

Put these in order:

A If the dam opens, the river rises.
B If the river rises, the ferry stops.
C The dam opens.
D So the ferry stops.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q4
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Our new ads worked: sales rose this month." Which assumption is the argument making?

A. Sales rose this month.
B. Nothing other than the ads caused most of the rise in sales.
C. Sales will rise next month.
D. The ads were expensive.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q5
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Set 4: Eliminate candidates clue by clue

Local outcome LOGIC.CONSTRAINT.01: Find the only candidate left after applying every clue to a finite list

Puzzle set · genre: elimination · MP1, MP6

MAIN: Find the only candidate left after every clue

BACKWARD 1: Strategy: cross out, one clue at a time

FORWARD 1: ADVANCED APPLICATION: Eliminate with clues about neighbors and positions

FORWARD 2: EXPERT CHALLENGE: Which clues are really needed?

Which month is it? Its name has fewer than 8 letters. Its name contains the letter R. It is the 1st, 3rd, 5th, 7th, 9th or 11th month. Its name does not start with J.

A. March
B. May
C. January
D. September

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q1
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Which day of the week is it? Its name has no letter E. It comes after Wednesday in the week that starts on Monday. Its name has exactly 6 letters. Its name contains the letter U.

A. Friday
B. Thursday
C. Sunday
D. Saturday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q2
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A whole number from 1 to 50 leaves remainder 3 when divided by 7. It is greater than 20. It is even. It does not leave remainder 4 when divided by 5. What is the number?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q3
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Three friends, Ana, Ben and Cai, each have exactly one pet: a cat, a dog or a fish, all different. Ana does not have the dog. Ben has neither the cat nor the dog. Which pet does each friend have?

Choose one column for each row. Each column is used once.

	cat	dog	fish
Ana			
Ben			
Cai			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q4
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A locker number is one of 12, 15, 18, 21, 24, 27. It is even. It is not a multiple of 4. Which locker is it?

A. 24
B. 27
C. 12
D. 18

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q5
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A number from 1 to 10 is even, greater than 7 and less than 10. What is it?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q1
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A day of the week (Monday to Sunday) starts with T and has more than 7 letters. Which day?

A. Saturday
B. Thursday
C. Tuesday
D. Wednesday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q2
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

How many months have a name with exactly 4 letters?

Months: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q3
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

If two candidates are left after every clue, the clues determine the answer.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q4
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates: 2, 3, 4, 5, 6. Clue 1: the number is prime. Clue 2: the number is odd. How many candidates are left?

A. 0
B. 3
C. 2
D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q5
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Which month is it? The month before it has a name longer than 5 letters. The month after it does not end in "ber". The month after it contains the letter U. Its own name contains the letter A.

A. January
B. May
C. February
D. December

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q1
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

In the week Monday to Sunday, exactly one day has a name whose day before and day after both contain the letter T.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q2
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A whole number from 1 to 100 is a perfect square, is odd, has two digits, and its digits add to 13. What is it?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q3
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Four runners, Dev, Eva, Finn and Gia, finished 1st to 4th, no ties. Eva finished right after Dev. Gia finished before Finn. Finn finished before Dev. Assign each runner a place.

Choose one column for each row. Each column is used once.

	1st	2nd	3rd	4th
Dev				
Eva				
Finn				
Gia				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q4
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates are the whole numbers 1 to 30. Clues: A: the number is odd. B: it is a multiple of 3. C: it is greater than 10. D: it is a multiple of 5. Which number fits all four clues?

A. 30
B. 15
C. 5
D. 25

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q5
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates are the whole numbers 1 to 30, with four clues: A: odd; B: a multiple of 3; C: greater than 10; D: a multiple of 5. Only 15 fits all four. Which ONE clue can be removed so that exactly one number is still left?

A. C (greater than 10)
B. D (multiple of 5)
C. A (odd)
D. B (multiple of 3)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q1
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A number from 1 to 30 is a multiple of 4. What is the least number of extra yes/no questions of the form "Is it greater than k?" that always identify it?

Questions: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q2
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Adding a new clue to a puzzle can never increase the number of candidates left.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q3
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates are the whole numbers 1 to 20. Write a number n that survives BOTH clues: n is prime, and n leaves remainder 1 when divided by 4.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q4
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Six boxes are numbered 1 to 6. One holds a prize. Which single clue, added alone, would leave exactly one box?

A. "The box number is odd."
B. "The box number is a multiple of 3."
C. "The box number is less than 4."
D. "The box number is prime and even."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q5
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Set 5: Negation and truth values

Local outcome LOGIC.CONNECTIVE.01: Form the negation of a statement and find truth values of simple statements

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Form negations and find truth values

BACKWARD 1: Truth values of simple statements

BACKWARD 2: Negation is not the same as the opposite

FORWARD 1: ADVANCED APPLICATION: Negate ranges, counts and double negatives

FORWARD 2: EXPERT CHALLENGE: Negation inside conditions and counts

What is the negation of "Sam is taller than Ana"?

A. Ana is taller than Sam.
B. Sam and Ana are the same height.
C. Sam is not taller than Ana.
D. Sam is shorter than Ana.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q1
Logic, Set 5: Negation and truth values · MAIN · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

x stands for a real number. What is the negation of "x > 5"?

A. $x \leq 5$

B. $x < 5$

C. $x \geq 5$

D. $x < -5$

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q2
Logic, Set 5: Negation and truth values · MAIN · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

P is the statement "48 is a multiple of 9." The negation of P is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q3
Logic, Set 5: Negation and truth values · MAIN · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Give a number x that makes the NEGATION of the statement "$x \leq 4$" true.

$x =$ _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q4
Logic, Set 5: Negation and truth values · MAIN · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Which pair of statements are negations of each other (in every situation, exactly one of the two is true)?

A. "The cup is full." / "The cup is empty."

B. "Every student passed." / "Every student failed."

C. "The coin landed heads up." / "The coin did not land heads up."

D. "x > 3" / "x < 3"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q5
Logic, Set 5: Negation and truth values · MAIN · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

"17 is a prime number" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q1
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

"3 x 7 = 24" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q2
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

"Every even number is a multiple of 4" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q3
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

Which statement is false?

A. 0.5 equals one half.
B. 0.5 + 0.25 = 0.75.
C. 0.75 is greater than 0.5.
D. 0.5 is greater than 0.75.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q4
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

"10 is not an even number" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q5
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

What is the negation of "The door is open"?

A. The door is locked.
B. The door is not open.
C. The door is wide open.
D. The window is open.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q1
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

"The water is cold" and "The water is hot" are negations of each other.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q2
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "Tom won the race"?

A. Tom did not win the race.
B. Someone else won.
C. Tom lost the race.
D. Tom came last.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q3
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

If a statement is true, its negation is also true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q4
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

n is a whole number. What is the negation of "n is even"?

A. n is 0.
B. n is negative.
C. n is odd.
D. n is prime.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q5
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

x stands for a real number. What is the negation of "3 < x < 7" (x is between 3 and 7)?

A. $x < 3$ or $x > 7$

B. $3 \geq x \geq 7$

C. $x \leq 3$ and $x \geq 7$

D. $x \leq 3$ or $x \geq 7$

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q1
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Exactly 9 lamps are in a room. What is the negation of "At least 5 lamps are on"?

A. At most 5 lamps are on.
B. At most 4 lamps are on.
C. No lamps are on.
D. At least 4 lamps are off.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q2
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Complete the truth table: the columns are "not P" and "not (not P)".

Fill in T or F in every empty cell.

P	not P	not (not P)
T		
F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q3
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

How many of these are true? (1) "It is not true that 8 is odd." (2) "It is false that 0 is not a whole number." (3) "It is not the case that 2 + 2 is not 4." (4) "It is not true that 5 is not less than 3."

Number true: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q4
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

The negation of "Mia always walks to school" is "Mia never walks to school".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q5
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Give a number x that makes the negation of "x < 2 or x > 5" true.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q1
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

There are 6 lights. What is the negation of "At most 2 lights are on"?

A. At least 2 lights are on.
B. At most 2 lights are off.
C. At least 3 lights are on.
D. Exactly 3 lights are on.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q2
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Ana says: "It is false that it is not true that the test is not on Friday." What is Ana saying?

A. Nothing about the test.
B. The test might be on Friday.
C. The test is on Friday.
D. The test is not on Friday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q3
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

For every statement P, exactly one of P and "not P" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q4
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Statements about a whole number n from 1 to 10: A: "n is even." B: "n is not greater than 6." C: "It is not true that n is a multiple of 3." For how many values of n are all three statements true?

Values: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q5
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Set 6: Crack a code with a stated key

Local outcome LOGIC.CODE.01: Decode and encode with a stated symbol or value key, and infer a key from examples within a stated family

Puzzle set · genre: symbol codes · MP7, MP2

- MAIN: Decode and encode with a stated key, and find a key within a stated family
- BACKWARD 1: Strategy: write the key as a table
- FORWARD 1: ADVANCED APPLICATION: Codes with structure: shapes of letters and combined rules
- FORWARD 2: EXPERT CHALLENGE: Find a key and prove when none exists

In a code, each letter has the value of its position in the alphabet (A = 1, B = 2, ..., Z = 26), and a word's value is the sum of its letters' values. For example, CAB = 3 + 1 + 2 = 6. What is the value of BEAD?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q1
Logic, Set 6: Crack a code with a stated key · MAIN · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In a code, every letter is replaced by the letter 3 places later in the alphabet (A becomes D, B becomes E, and X, Y, Z wrap around to A, B, C). A coded word reads FDW. What was the original word?

A. CAW
B. CAT
C. DBX
D. IGZ

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q2
Logic, Set 6: Crack a code with a stated key · MAIN · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code adds the same whole number k to every digit and keeps only the last digit of each sum (so 8 + 3 gives 1). The code turns 2025 into 4247. What does it turn 1990 into?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q3
Logic, Set 6: Crack a code with a stated key · MAIN · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Each symbol stands for a whole number. Star + Star = 10. Star + Circle = 8. Circle + Square = 9. What number is Square?

Square: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q4
Logic, Set 6: Crack a code with a stated key · MAIN · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Two 3-letter words were interleaved: first letter of word 1, first letter of word 2, second letter of word 1, and so on. The result is CDAORT. Which two words were interleaved?

A. CAR and DOT
B. CAD and ORT
C. COD and ART
D. CAT and DOR

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q5
Logic, Set 6: Crack a code with a stated key · MAIN · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code replaces each letter by its position in the alphabet (A = 1, ..., Z = 26) and adds them. What is the value of DOG?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q1
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code moves each letter 1 place later (A -> B, ..., Z -> A). What is BOX in code?

A. CPY
B. DQZ
C. CPX
D. ANW

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q2
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code replaces each digit d by 9 - d. What is the code for 306?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q3
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

If a code moves every letter 2 places later, then decoding moves every letter 2 places earlier.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q4
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Triangle + Triangle + Triangle = 12. What number is Triangle?

Triangle: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q5
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. What is the value of BREAD?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q1
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Which word has value 0?

A. SHIRT
B. FLIES
C. MAZE
D. TOWEL

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q2
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code first writes a word backward and then moves each letter 1 place later in the alphabet. What is the code for MAP?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q3
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Each symbol stands for a whole number. Moon + Moon + Sun = 13. Sun + Sun = 6. Moon + Star = 9. What number is Star?

Star: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q4
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code moves every letter 13 places later in the alphabet (wrapping around). Applying this code twice gives back the original word.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q5
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code adds 3 to each digit of a 4-digit number and keeps only the last digit of each sum (so 8 becomes 1). Is there a 4-digit number whose code is the same number written backward?

A. No: but only because the first digit cannot be 0.

B. Yes: 1234 works.

C. No: the middle digits would need b + 6 to end in the same digit as b, which never happens.

D. Yes: any number whose digits are all equal works.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q1
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code moves every letter the same number of places later in the alphabet (wrapping around). It turns GO into MU. What does it turn LOGIC into?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q2
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Each symbol stands for a whole number from 1 to 9. Square + Square = Circle, and Circle + Square = 12. Find Square s and Circle c.

Square s = _____

Circle c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q3
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Using only the letters B, O and T (each as often as you like), what is the greatest value of a 4-letter string?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q4
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code sends each letter to the letter at the same distance from the other end of the alphabet (A <-> Z, B <-> Y, C <-> X, ...). Which word is coded as itself?

A. WIZARD

B. MOON

C. NOON

D. No word: no letter is coded as itself.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q5
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Set 7: Order from relational clues

Local outcome LOGIC.ORDER.01: Build an ordering that satisfies relational clues (before, after, between, next to)

Puzzle set · genre: ordering · MP1, MP2

MAIN: Build an ordering that satisfies relational clues

BACKWARD 1: Strategy: place the most constrained item first

FORWARD 1: ADVANCED APPLICATION: Orderings with "next to" and "not next to"

FORWARD 2: EXPERT CHALLENGE: Count and compare orderings

Four books stand on a shelf. The atlas is left of the cookbook. The diary is right of the cookbook. The biography is at the far left. Order the books from left to right.

Put these in order:

A atlas
B biography
C cookbook
D diary

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q1
Logic, Set 7: Order from relational clues · MAIN · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five friends arrived at a party one at a time. Gil arrived before Hana but after Ivo. Jo arrived last. Kai arrived right after Ivo. Order the arrivals, first to last.

Put these in order:

G Gil
H Hana
I Ivo
J Jo
K Kai

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q2
Logic, Set 7: Order from relational clues · MAIN · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five runners finished a race with no ties. Uma beat Vic. Wes finished right after Uma. Xia beat Uma. Yan finished last. Who finished in the middle (3rd)?

A. Uma
B. Vic
C. Wes
D. Xia

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q3
Logic, Set 7: Order from relational clues · MAIN · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four flags hang on a pole, from top to bottom. The green flag is somewhere between the red and the blue flags. The red flag is not at the top. The yellow flag is directly below the blue flag. Order the flags from top to bottom.

Put these in order:

R red
G green
B blue
Y yellow

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q4
Logic, Set 7: Order from relational clues · MAIN · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Seven students stand in a line. Lee is 2nd from the left. Max is 3rd from the right. How many students stand between Lee and Max?

Students between: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q5
Logic, Set 7: Order from relational clues · MAIN · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Three cups stand in a row. The red cup is between the other two. The blue cup is on the right. Order them from left to right.

Put these in order:

R red
B blue
W white

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q1
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

In a line of 9 people, Ola is 4th from the front. What is her position counted from the back?

Position from the back: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q2
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

"Bea is right after Cy" means no one is between Cy and Bea.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q3
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Ava is taller than Ben, and Ben is taller than Cal. Who is shortest?

A. Ben
B. Cal
C. Ava
D. It cannot be known.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q4
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four racers: Dot finished first. Eli beat Fay. Gus finished last. Order them, first to last.

Put these in order:

D Dot
E Eli
F Fay
G Gus

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q5
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five houses stand in a row, numbered 1 to 5 from left to right. The bakery is next to the bank. The café is at one end. The dentist is directly right of the café. The florist is not next to the dentist. The bank is in house 4. Order the businesses from left to right.

Put these in order:

K bakery
N bank
C café
D dentist
F florist

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q1
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Six people, A to F, sit in seats 1 to 6 in a row. C is in seat 1 and D is in seat 6. B sits right after A, and E sits right after B. F does not sit next to C. Which seat is F in?

Seat: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q2
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five students line up. Pat is in front of Quinn. Rae is right behind Sol. Tia is first. Quinn is last. Which statement must be true?

A. Rae is 4th.

B. Pat is 2nd.

C. Sol is right behind Pat.

D. Tia is directly in front of Pat or of Sol.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q3
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four friends sit in seats 1 to 4 in a row. Ari is not next to Bo. Cy sits in seat 2. Bo is in a higher seat number than Cy. Di is not in seat 4. Give each friend's seat.

Choose one column for each row. Each column is used once.

	1	2	3	4
Ari				
Bo				
Cy				
Di				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q4
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

If five people stand in a line and three of them must stand next to each other in a fixed order, there are exactly 6 possible lines.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q5
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five students, P, Q, R, S and T, stand in a line. P is somewhere in front of Q, and R is somewhere in front of S. How many different lines are possible?

Lines: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q1
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Six relay swimmers, A to F, swim in order. B swims right after E. D swims before A. C swims last. F swims first. A swims right after B. Give the swimming order.

Put these in order:

A A
B B
C C
D D
E E
F F

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q2
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four people, W, X, Y and Z, stand in a line. Clues: (1) W is not first. (2) X is right after W. (3) Y is last. Which clue could be removed and still leave exactly one possible line?

A. No clue can be removed.

B. Clue 3

C. Clue 1

D. Clue 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q3
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Seven people stand in a line, numbered 1 to 7 from the front. Ana is in position a and Ben is in position b. Ana is in front of Ben, exactly two people stand between them, and Ben is not last. Give one possible pair of positions.

Ana a = _____

Ben b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q4
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

With the clues "J is before K" and "K is before L" for three people in a line, the order is fully determined.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q5
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Set 8: And, inclusive or, exclusive or

Local outcome LOGIC.CONNECTIVE.02: Find truth values of conjunctions and of inclusive and exclusive disjunctions

Formal logic set · genre: formal logic · MP6

- MAIN: Truth values of and, inclusive or, exclusive or
- BACKWARD 1: Truth values of the parts
- BACKWARD 2: Inclusive or exclusive?
- FORWARD 1: ADVANCED APPLICATION: Compound statements with three parts
- FORWARD 2: EXPERT CHALLENGE: Design statements with given truth values

"7 is odd and 7 is greater than 10" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q1
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"7 is odd or 7 is greater than 10" is true (with "or" meaning at least one).

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q2
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

A sign says: "Each meal comes with soup or salad, but not both." Which order follows the sign?

A. a meal with salad and no soup
B. a meal with soup and salad
C. a meal with two salads and a soup
D. a meal with neither soup nor salad

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q3
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Give a whole number n that makes "n is even or n is greater than 20" true but "n is even and n is greater than 20" false.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q4
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

P is true and Q is false. Which statement is FALSE?

A. P and not Q
B. P exclusive-or Q
C. P or Q
D. P and Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q5
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"12 is a multiple of 5" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q1
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

"-3 < 1" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q2
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

"A triangle has 4 sides" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q3
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

"0 is even" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q4
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

Which part of "9 is odd and 9 is prime" is false?

A. both parts
B. 9 is prime
C. 9 is odd
D. neither part

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q5
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

With inclusive "or", "P or Q" is false when both P and Q are true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q1
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

With exclusive "or", "P or Q" is true when both P and Q are true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q2
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

A rule says "You may bring a pencil or a pen (or both)." Which kind of "or" is this?

A. neither kind

B. it cannot be told

C. inclusive

D. exclusive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q3
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"The light is either on or off" describes two cases that can never both happen. Which "or" fits best?

A. "and"
B. exclusive
C. inclusive
D. neither

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q4
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

When P is false and Q is false, exclusive "P or Q" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q5
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"(3 > 2 and 5 > 9) or 4 is even" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q1
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"3 > 2 and (5 > 9 or 4 is odd)" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q2
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

For how many whole numbers n from 1 to 20 is "n is a multiple of 3 exclusive-or n is a multiple of 4" true?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q3
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Complete the truth table for "P exclusive-or Q" and for "P or Q" (inclusive).

Fill in T or F in every empty cell.

P	Q	P exclusive-or Q	P or Q
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q4
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Statement S: "P or Q" (inclusive). S is false. What can be concluded?

A. P is false and Q is false.
B. P is true and Q is false.
C. P is false or Q is false.
D. Exactly one of P and Q is false.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q5
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

P, Q and R are statements. In how many of the 8 combinations of truth values is "(P or Q) and not R" true?

Combinations: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q1
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Give whole numbers a and b with "a > b exclusive-or a + b is even" true and both numbers between 1 and 9.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q2
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"P exclusive-or Q" has the same truth value as "(P or Q) and not (P and Q)" in every row of a truth table.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q3
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Three lamps A, B and C are each on or off. Exactly one of these statements is true: (1) "A is on or B is on." (2) "C is on." (3) "A is on exclusive-or C is on." Which lamps are on?

A. A only
B. A and B
C. none
D. B only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q4
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

For how many whole numbers n from 1 to 30 is "n is odd and (n is a multiple of 3 or n is a multiple of 5)" true?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q5
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

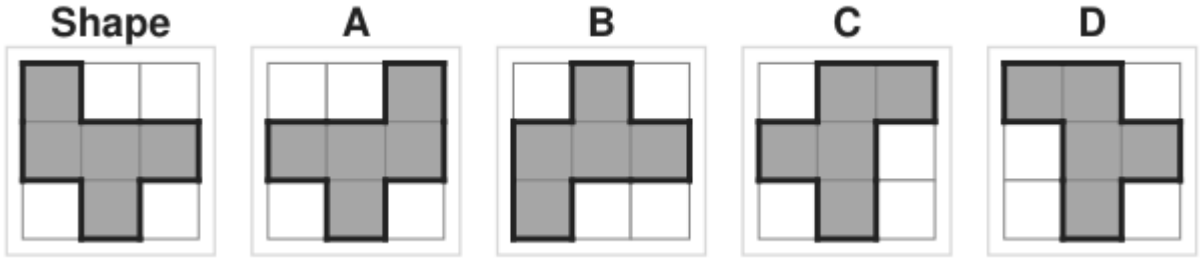
Set 9: Rotation or reflection?

Local outcome LOGIC.SPATIAL.02: Distinguish rotations from reflections of a figure on a grid

Puzzle set · genre: rotation and reflection · MP7

- MAIN: Distinguish rotations from reflections
- BACKWARD 1: Strategy: follow one marked corner
- FORWARD 1: ADVANCED APPLICATION: Combine turns and flips
- FORWARD 2: EXPERT CHALLENGE: Symmetry under turns and flips

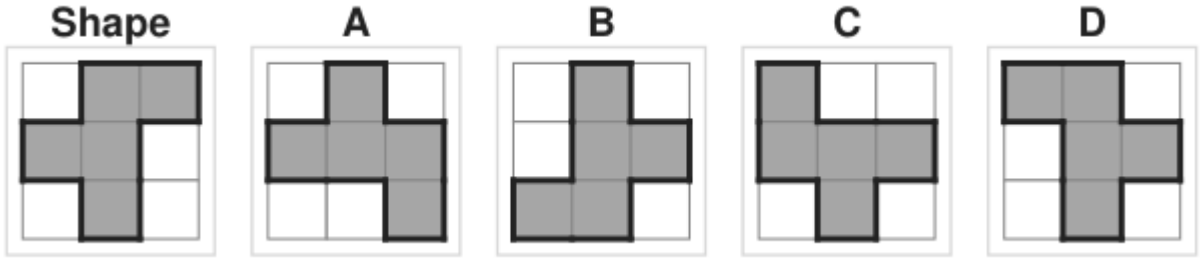
Which figure shows the shape TURNED (rotated in the page), not flipped over? (Only the shape matters, not where it sits in its box.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q1
Logic, Set 9: Rotation or reflection? · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

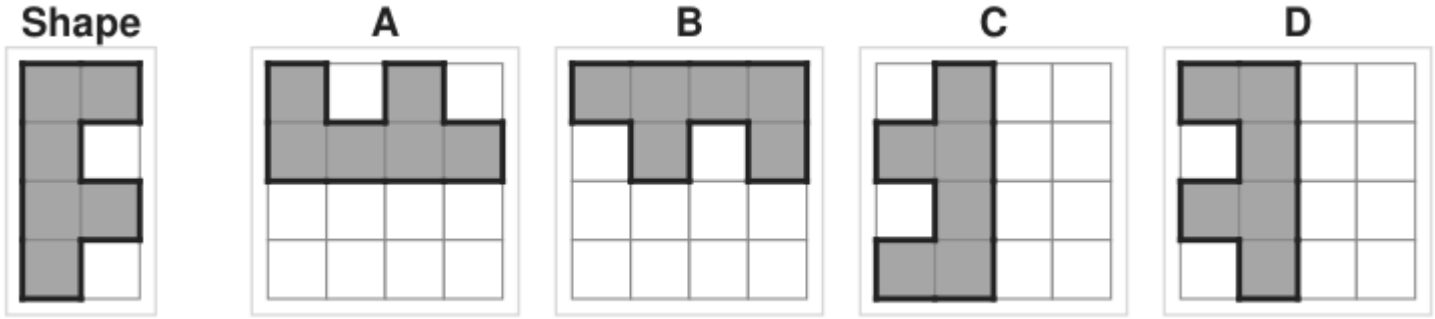
Three of the figures are the shape turned (rotated). One is the shape flipped over (a mirror image). Which figure is the flipped one?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q2
Logic, Set 9: Rotation or reflection? · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

The shape is turned a quarter-turn CLOCKWISE (90 degrees, like a clock hand moving from 12 to 3). Which figure shows the result?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

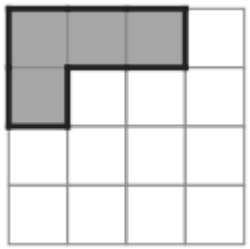
REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q3
Logic, Set 9: Rotation or reflection? · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Capital letters are turned upside down by a half-turn (180 degrees in the page). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. Which letter looks the same after the half-turn?

A. A
B. M
C. N
D. E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q4
Logic, Set 9: Rotation or reflection? · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A 4 by 4 grid has shaded cells at (1,1), (1,2), (1,3) and (2,1), written as row,column with row 1 at the top. The whole grid is turned a quarter-turn clockwise about its center. Mark the cells where the shaded cells end up.



Mark cells on the 4 × 4 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q5
Logic, Set 9: Rotation or reflection? · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A quarter-turn clockwise followed by another quarter-turn clockwise is the same as a half-turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q1
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

An arrow points up. It is turned a quarter-turn clockwise. Which way does it point now?

A. right
B. up
C. left
D. down

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q2
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

An arrow points right. It is reflected in a vertical mirror line. Which way does it point now?

A. down
B. left
C. right
D. up

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q3
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Flipping a shape over twice in the same mirror line brings it back to where it started.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q4
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

How many quarter-turns clockwise bring a shape back to its starting position for the first time, if the shape has no symmetry at all?

Quarter-turns: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q5
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Reflecting a shape in a vertical line and then in a horizontal line gives the same result as a half-turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q1
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A shape is turned a quarter-turn clockwise three times. Which single move gives the same result?

A. a reflection
B. a quarter-turn clockwise
C. a quarter-turn counterclockwise
D. a half-turn

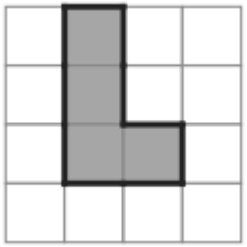
REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q2
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A word written in capital letters is turned a half-turn (180 degrees). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. A half-turn also reverses the order of the letters. Which word looks exactly the same after the turn?

A. NOW
B. MOM
C. ZOO
D. SOS

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q3
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A 4 by 4 grid has shaded cells at (1,2), (2,2), (3,2) and (3,3), written as row,column with row 1 at the top. The grid is given a half-turn about its center. Mark where the shaded cells end up.



Mark cells on the 4 × 4 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q4
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A shape has no symmetry. How many different-looking positions can it be put in using turns AND flips (on a grid, counting each distinct picture once)?

Positions: _____

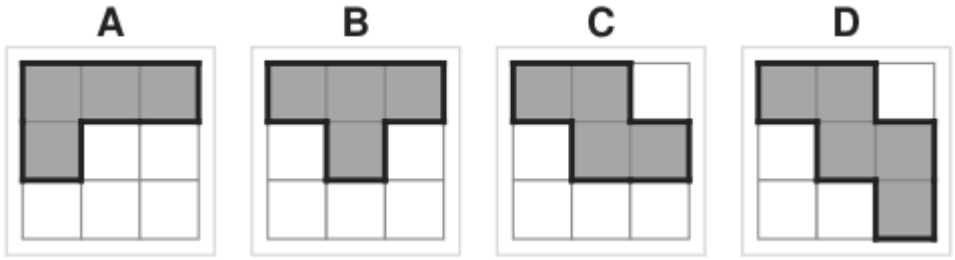
REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q5
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Of the 8 moves of a square grid figure (4 turns: 0, 90, 180, 270 degrees; and 4 flips: vertical, horizontal and both diagonals), how many leave a plus-shaped figure of 5 cells (a center cell with one cell on each side) unchanged?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q1
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

For which figure can the mirror image be produced by turning the figure (without flipping it over)? Exactly two of the four figures have this property; choose the answer that names both.



A. B and D
B. A and C
C. A and B
D. C and D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q2
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

An S-shaped figure of 4 cells (two cells in a row, and two more cells one row down, shifted one place left) is turned and flipped. Which statement is true?

A. No move other than doing nothing leaves it unchanged.

B. A quarter-turn leaves it unchanged.

C. A half-turn leaves it unchanged, but no flip does.

D. A flip in a vertical line leaves it unchanged.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q3
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

If a figure looks the same after a quarter-turn, then it also looks the same after a half-turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q4
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

How many of the 7 shapes made of 4 grid cells joined edge to edge (counting a shape and its turns as the same, but its mirror image as different) look the same after a half-turn?

Shapes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q5
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Set 10: Deduction and induction

Local outcome LOGIC.ARGUMENT.02: Tell deductive conclusions (forced by the premises) from inductive ones (supported by examples)

Formal logic set · genre: formal logic · MP3, MP8

- MAIN: Tell deductive conclusions from inductive ones
- BACKWARD 1: Premises and conclusions
- BACKWARD 2: Certain or only likely?
- FORWARD 1: ADVANCED APPLICATION: Patterns that break
- FORWARD 2: EXPERT CHALLENGE: From pattern to proof

Which argument is DEDUCTIVE (the conclusion must be true if the premises are true)?

A. Every swan I have seen is white. So all swans are white.

B. Every prime number greater than 2 is odd. 31 is a prime number greater than 2. So 31 is odd.

C. Most students like recess. So Ana likes recess.

D. The last 20 buses were on time. So the next bus will be on time.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q1
Logic, Set 10: Deduction and induction · MAIN · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16. So the sum of the first n odd numbers is always n x n." What kind of reasoning is this?

A. deductive: the examples prove it for every n

B. neither: it is not an argument

C. deductive: the conclusion is true

D. inductive: it generalizes from a few examples

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q2
Logic, Set 10: Deduction and induction · MAIN · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

In an inductive argument, the premises can all be true while the conclusion is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q3
Logic, Set 10: Deduction and induction · MAIN · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

n x n + n + 11 is prime for n = 1, 2, 3, ..., 9. A student concludes it is prime for every whole number n. Which value of n shows the conclusion is false?

A. 10

B. 8

C. 4

D. 9

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q4
Logic, Set 10: Deduction and induction · MAIN · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which conclusion follows DEDUCTIVELY from "All members of the chess club are in grade 10 or grade 11, and Jun is a member of the chess club"?

A. Jun is good at chess.

B. Jun is in grade 10 or grade 11.

C. Jun is in grade 11.

D. Most grade 10 students are in the chess club.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q5
Logic, Set 10: Deduction and induction · MAIN · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"All the cookies are gone, so someone ate them." Which part is the conclusion?

A. nobody ate them
B. all the cookies are gone
C. cookies are tasty
D. someone ate them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q1
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

A premise is the statement that an argument is trying to prove.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q2
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

"It rained every day this week. So it will rain tomorrow." What is the premise?

A. It rained every day this week.
B. It will rain tomorrow.
C. It never rains.
D. Rain is good for plants.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q3
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

Every argument has at least one premise and one conclusion.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q4
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

"Since 12 is even, 12 is divisible by 2." What is the conclusion?

A. 12 is even
B. 12 is odd
C. 12 is divisible by 2
D. 2 is even

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q5
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

"Most tickets in the box are red, and I drew a ticket from the box" makes "my ticket is red" certain.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q1
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"Most tickets in the box are red, and I drew a ticket" makes "my ticket is red" certain.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q2
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which word signals that a conclusion is only likely?

A. must
B. probably
C. every
D. therefore

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q3
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

If a pattern holds for 100 examples, it must hold for the 101st.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q4
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

A die has been rolled 5 times and showed an even number each time. Which conclusion is certain?

A. The die is unfair.

B. The next roll will be even.

C. Each of those 5 rolls was even.

D. The die only has even numbers.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q5
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

The claim "n x n - n + 17 is prime for every whole number n" holds for n = 1 to 16. Find a whole number n that makes n x n - n + 17 NOT prime.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q1
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Points are placed on a circle and every pair is joined by a straight line. With 1, 2, 3, 4, 5 points (in general position), the circle is cut into 1, 2, 4, 8, 16 regions. A student predicts 32 regions for 6 points. What is the actual number?

A. 31
B. 32
C. 36
D. 30

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q2
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

A deductive argument with true premises can have a false conclusion.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q3
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which statement is the strongest inductive argument?

A. All bolts in the batch passed, so this bolt passed.
B. My two friends like the song, so everyone does.
C. The first bolt passed, so all bolts pass.
D. All 500 randomly chosen bolts from the batch passed, so nearly all bolts in the batch pass.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q4
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which reasoning is deductive?

A. Most rectangles in the book are squares, so this one is.

B. All squares are rectangles; this figure is a square; so it is a rectangle.

C. Every rectangle I measured had equal diagonals, so all rectangles do.

D. This figure looks like a rectangle, so it probably is one.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q5
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

The expression n x n + n + 41 is prime for n = 0, 1, 2, ..., 39. What is the smallest whole number n for which it is NOT prime?

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q1
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"For every whole number n, n x n - n is even, because 2 x 2 - 2 = 2, 3 x 3 - 3 = 6 and 4 x 4 - 4 = 12 are even." Which statement about this argument is correct?

A. The claim is true, but three examples do not prove it.

B. The claim is false, because 1 x 1 - 1 = 0.

C. The argument proves the claim, because three cases are checked.

D. The claim is false for large n.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q2
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

If every case of a finite list has been checked, the conclusion about that list is deductive.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q3
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Give a whole number n from 1 to 20 for which "2 to the power n, minus 1, is prime" is FALSE, even though n is prime. (2 to the power n is 2 multiplied by itself n times.)

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q4
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

A student checks that every whole number from 4 to 100 that is even can be written as a sum of two primes. Which conclusion is justified?

A. Every even number greater than 2 is a sum of two primes.

B. Every even number is a sum of two primes.

C. Every even number from 4 to 100 is a sum of two primes.

D. No odd number is a sum of two primes.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q5
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Set 11: The structure of words

Local outcome LOGIC.WORD.01: Use letter counts, positions and alphabetical order to test word patterns within a stated rule

Puzzle set · genre: word structure · MP7, MP6

MAIN: Use letter counts, positions and order to test word patterns
BACKWARD 1: Strategy: list the letters with their positions
FORWARD 1: ADVANCED APPLICATION: Patterns across several words
FORWARD 2: EXPERT CHALLENGE: Count and construct letter strings under rules

Which word has its letters in alphabetical order (each letter the same as or later in the alphabet than the one before)?

A. CHART
B. BEGAN
C. FLOWER
D. ALMOST

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q1
Logic, Set 11: The structure of words · MAIN · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many times does the letter S appear in the word ASSESSMENTS?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q2
Logic, Set 11: The structure of words · MAIN · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Number words are written in capital letters (ONE, TWO, ..., TEN). Which number word has exactly as many letters as the number it names?

A. FOUR
B. THREE
C. FIVE
D. TWO

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q3
Logic, Set 11: The structure of words · MAIN · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Take the first letter of each word: "Every Apple Tastes Sweet." What word do the first letters spell?

A. TEAS
B. EAST
C. EATS
D. SEAT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q4
Logic, Set 11: The structure of words · MAIN · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Write the letters of STEAM in alphabetical order. Which letter is in the middle position?

A. A
B. M
C. S
D. E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q5
Logic, Set 11: The structure of words · MAIN · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

The letters of BELLOW are in alphabetical order.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q1
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many times does the letter E appear in REFERENCE?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q2
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

What is the 4th letter of the word PLANET?

A. E
B. A
C. L
D. N

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q3
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Take the first letter of each word: "Owls Watch Lakes." What do the first letters spell?

A. OLW
B. OWL
C. WOL
D. LOW

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q4
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many letters does the word EIGHT have?

Letters: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q5
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

In the list GO, CAT, LAMP, PLANT, each word has one more letter than the word before. Which word could come next?

A. BALLOON
B. TREE
C. GARDEN
D. HOUSE

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q1
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many of the number words ONE, TWO, THREE, ..., TEN have their letters in alphabetical order?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q2
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Which word has its letters in strictly reverse alphabetical order (each letter earlier in the alphabet than the one before it)?

A. SPELL
B. TRACK
C. STONE
D. SPOKE

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q3
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Write the letters of each number word ONE to TEN in alphabetical order; for example, ONE becomes ENO. Which number word becomes EENSV? Give its value as a number.

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q4
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Every word whose letters are in alphabetical order reads differently backward unless all its letters are the same.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q5
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many 3-letter strings can be made from the letters A, B, C, D (letters may repeat) whose letters are in alphabetical order (each letter the same as or later than the one before)?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q1
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many 4-letter strings using only the letters X and Y have no two equal letters next to each other?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q2
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many 4-letter strings using the letters A, B, C (letters may repeat) have no letter next to itself?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q3
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

A 6-letter string uses each of the letters A, B, C exactly twice, and no letter is next to itself. Which string follows the rule?

A. ABCABC
B. AABBCC
C. ABBCAC
D. ACBBCA

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q4
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

The five letters of LEVEL (L, E, V, E, L) are rearranged into 5-letter strings. How many different strings have no two equal letters next to each other?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q5
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Set 12: When is "if ... then" false?

Local outcome LOGIC.CONDITIONAL.01: Find the truth value of a conditional and identify the one case that makes it false

Formal logic set · genre: formal logic · MP3, MP6

MAIN: Find when a conditional is true and the one case that makes it false
BACKWARD 1: Find the "if" part and the "then" part
BACKWARD 2: Truth values of the parts
FORWARD 1: ADVANCED APPLICATION: Conditionals in mathematics
FORWARD 2: EXPERT CHALLENGE: Nested and chained conditionals

A rule: "If a package weighs more than 5 kg, then it has a yellow label." Which package BREAKS the rule?

A. a 7 kg package with a yellow label
B. a 7 kg package with a blue label
C. a 3 kg package with a blue label
D. a 3 kg package with a yellow label

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q1
Logic, Set 12: When is "if ... then" false? · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Complete the truth table for "if P then Q".

Fill in T or F in every empty cell.

P	Q	if P then Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q2
Logic, Set 12: When is "if ... then" false? · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Find a whole number n that is a counterexample to "If n is odd, then n is prime." (A counterexample makes the "if" part true and the "then" part false.)

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q3
Logic, Set 12: When is "if ... then" false? · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If 5 > 8, then 2 + 2 = 5" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q4
Logic, Set 12: When is "if ... then" false? · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If it is a weekend day, the library opens at noon." Today the library opened at 9 a.m. What follows?

A. The rule was broken.
B. Today is a weekend day.
C. Today is not a weekend day.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q5
Logic, Set 12: When is "if ... then" false? · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "If the sun sets, the street lights turn on," what is the "if" part (the hypothesis)?

A. the street lights turn on
B. the lights are bright
C. it is night
D. the sun sets

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q1
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "Students may enter the lab only if they wear goggles," which is the "then" part when it is written as "If ..., then ..."?

A. they wear goggles
B. students enter the lab
C. they leave the lab
D. the lab is open

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q2
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Rewrite "Every multiple of 10 ends in 0" as a conditional.

A. If a number is not a multiple of 10, then it does not end in 0.
B. If a number ends in 0, then it is a multiple of 10.
C. If a number is a multiple of 10, then it ends in 0.
D. A number ends in 0 and is a multiple of 10.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q3
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "If x = 3, then x + 1 = 4," the hypothesis is "x + 1 = 4".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q4
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "The alarm rings whenever the door opens," which part is the hypothesis?

A. the door opens
B. the alarm rings
C. the door closes
D. the alarm is loud

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q5
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"6 is a factor of 30" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q1
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

"4 squared is 8" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q2
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

"Every square is a triangle" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q3
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

Which statement is true?

A. -5 is greater than 2.
B. -5 is less than 2.
C. 2 is less than -5.
D. -5 equals 5.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q4
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

"1 is a prime number" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q5
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

Find a real number x that makes "If x squared is greater than 9, then x is greater than 3" FALSE.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q1
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If a whole number is divisible by 6, then it is divisible by 3" is true for every whole number.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q2
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

For how many whole numbers n from 1 to 12 is "If n is even, then n is a multiple of 4" FALSE?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q3
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

P is false. Which statement MUST be true?

A. If Q then P
B. P and Q
C. Q
D. If P then Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q4
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If you score above 90, you earn a badge." Lin earned a badge. What follows?

A. The rule was broken.
B. Lin scored 90 or below.
C. Nothing about Lin's score follows.
D. Lin scored above 90.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q5
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Complete the truth table for "if P then (if Q then R)".

Fill in T or F in every empty cell.

P	Q	R	if Q then R	if P then (if Q then R)
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q1
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In how many of the 8 rows of a truth table for P, Q, R is "(if P then Q) and (if Q then R)" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q2
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If P then (if Q then R)" has the same truth value as "if (P and Q) then R" in every row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q3
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If P then (if Q then R)" has the same truth value as "if (if P then Q) then R" in every row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q4
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Find whole numbers a and b that make the statement "If a + b is even, then a and b are both even" FALSE.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q5
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Set 13: Necessary and sufficient conditions

Local outcome LOGIC.CONDITIONAL.02: Translate "only if", "unless", "necessary" and "sufficient" into conditionals

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Translate "only if", "unless", "necessary" and "sufficient"
BACKWARD 1: Write a statement as "if ..., then ..."
BACKWARD 2: Necessary means required; sufficient means enough
FORWARD 1: ADVANCED APPLICATION: Combine conditions
FORWARD 2: EXPERT CHALLENGE: Chains of conditions

"Being at least 16 years old is NECESSARY for getting a driving permit in this town." Which conditional says the same thing?

A. If a person has a driving permit, then the person is at least 16.
B. If a person is at least 16, then the person has a driving permit.
C. If a person is under 16, then the person may still get a permit.
D. A person is at least 16 and has a permit.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q1
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"You may enter the museum only if you have a ticket." Which statement means the same?

A. If you have a ticket, then you may enter the museum.
B. You have a ticket and you enter.
C. If you have no ticket, then you may still enter.
D. If you enter the museum, then you have a ticket.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q2
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

In logic, "P unless Q" means "if not Q, then P". Which statement means the same as "The picnic is held unless it rains"?

A. If it rains, the picnic is held.
B. If it does not rain, the picnic is held.
C. If the picnic is held, it does not rain.
D. If it does not rain, the picnic is not held.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q3
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Being a square is a NECESSARY condition for being a rectangle.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q4
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

n is a whole number. Which condition is NECESSARY for "n is divisible by 6"?

A. n is divisible by 12
B. n is divisible by 9
C. n is even
D. n ends in the digit 6

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q5
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Rewrite "All birds have feathers" as a conditional.

A. If an animal is a bird, then it has feathers.
B. If an animal is not a bird, then it has no feathers.
C. Birds and feathers.
D. If an animal has feathers, then it is a bird.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q1
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

Rewrite "When the bell rings, class ends" as a conditional.

A. If class ends, then the bell rings.
B. The bell rings and class ends.
C. If the bell does not ring, class does not end.
D. If the bell rings, then class ends.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q2
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"If x = 2, then x + x = 4" has hypothesis "x + x = 4".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q3
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

What is the conclusion of "If a number ends in 5, it is divisible by 5"?

A. it is 5
B. it is divisible by 5
C. a number ends in 5
D. it is odd

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q4
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"Every even number is divisible by 2" can be written "If a number is even, then it is divisible by 2".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q5
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

Having fuel is the only thing a gas car needs in order to run.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q1
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Having fuel is sufficient for a gas car to run.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q2
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Which is SUFFICIENT for a whole number to be even?

A. It ends in 1.
B. It is a multiple of 10.
C. It is a multiple of 3.
D. It is greater than 2.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q3
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

If A is sufficient for B, then whenever B happens, A happens.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q4
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Being a mammal is ___ for being a dog.

A. necessary
B. both necessary and sufficient
C. sufficient
D. neither necessary nor sufficient

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q5
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"n is even" is necessary for "n is divisible by 6". Give a whole number n that shows it is NOT sufficient: n is even but not divisible by 6.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q1
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

n is a whole number. Which condition is SUFFICIENT for "n is divisible by 4"?

A. n is greater than 4
B. n is even
C. n is divisible by 2 and by 3
D. n is divisible by 8

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q2
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"A triangle is equilateral only if it is isosceles (has at least two equal sides)." Which statement is correct?

A. Being equilateral is necessary for being isosceles.
B. Being isosceles is sufficient for being equilateral.
C. Being isosceles is necessary for being equilateral.
D. The two conditions mean the same thing.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q3
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"You get the discount unless you pay late." Using "P unless Q" = "if not Q, then P", which person is guaranteed the discount?

A. no one
B. everyone
C. someone who paid late
D. someone who did not pay late

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q4
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

A condition can be both necessary and sufficient for another.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q5
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Facts: A is sufficient for B. B is sufficient for C. Which statement must be true?

A. C is necessary for A.
B. B is sufficient for A.
C. C is sufficient for A.
D. A is necessary for C.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q1
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

For whole numbers n from 1 to 60, condition X: "n is divisible by 4"; condition Y: "n is divisible by 6". How many n satisfy Y but not X (showing X is not necessary for Y)?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q2
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Which condition on a whole number n is BOTH necessary and sufficient for "n is divisible by 10"?

A. n is divisible by 5

B. n is divisible by 20

C. n is divisible by 2 and by 5

D. n is even

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q3
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

If A is necessary for B and B is necessary for A, then A and B are always both true or both false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q4
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Give a whole number n for which "n is a multiple of 3" holds but "n is a multiple of 9" fails, showing that being a multiple of 3 is not sufficient for being a multiple of 9.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q5
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Set 14: Logic grid puzzles

Local outcome LOGIC.GRID.01: Solve a matching puzzle with a grid by combining clues until one assignment remains

Puzzle set · genre: logic grids · MP1, MP6

MAIN: Solve a matching puzzle by combining clues
BACKWARD 1: Strategy: mark X for "no" and O for "yes" in a grid
FORWARD 1: ADVANCED APPLICATION: Grids with relative clues
FORWARD 2: EXPERT CHALLENGE: Two-attribute grids and decisive clues

Three students, Ama, Bex and Cho, each play one instrument: drums, flute or violin (all different). Ama does not play the flute. Bex plays neither the drums nor the flute. Which instrument does each student play?

Choose one column for each row. Each column is used once.

	drums	flute	violin
Ama			
Bex			
Cho			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q1
Logic, Set 14: Logic grid puzzles · MAIN · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four friends, Dara, Eli, Faye and Gus, each ordered a different drink: cocoa, juice, milk or tea. Dara did not order a hot drink (cocoa and tea are hot). Eli ordered tea. Faye ordered neither cocoa nor juice. Who ordered what?

Choose one column for each row. Each column is used once.

	cocoa	**juice**	**milk**	**tea**
Dara				
Eli				
Faye				
Gus				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q2
Logic, Set 14: Logic grid puzzles · MAIN · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four runners wore bibs 1, 2, 3 and 4 (all different): Hal, Ivy, Jun and Kit. Ivy's number is twice Hal's. Hal's number is even. Jun's number is odd. Kit's number is greater than Jun's. What number did each runner wear?

Choose one column for each row. Each column is used once.

	1	2	3	4
Hal				
Ivy				
Jun				
Kit				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q3
Logic, Set 14: Logic grid puzzles · MAIN · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three pets, a cat, a dog and a rabbit, belong to Lea, Max and Noa (one each). The cat's owner sits next to Max. Lea owns neither the dog nor the cat. Noa does not own the dog. Who owns which pet?

Choose one column for each row. Each column is used once.

	cat	dog	rabbit
Lea			
Max			
Noa			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q4
Logic, Set 14: Logic grid puzzles · MAIN · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four lockers, numbered 1 to 4 in a row, belong to Oz, Pia, Quin and Rae (one each). Pia's locker is directly to the right of Oz's (one number higher). Rae's locker is at an end of the row. Quin's locker number is even. Rae's number is lower than Quin's. Oz's locker is not number 2. Who has which locker?

Choose one column for each row. Each column is used once.

	1	2	3	4
Oz				
Pia				
Quin				
Rae				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q5
Logic, Set 14: Logic grid puzzles · MAIN · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Two friends, Ann and Bo, each pick a different color: red or blue. Ann does not pick red. Who picks which color?

Choose one column for each row. Each column is used once.

	red	blue
Ann		
Bo		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q1
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a 3 by 3 grid (3 people, 3 jobs, one each), a person's row has X in two cells. What follows?

A. Nothing follows yet.
B. The person has the job in the third cell.
C. The person has no job.
D. Another person has that job.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q2
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a matching puzzle where each option is used once, if Kim has the bike, no one else has the bike.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q3
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three cousins, Cy, Di and Ed, are ages 8, 9 and 10 (all different). Cy is the youngest. Ed is not 10. How old is each?

Choose one column for each row. Each column is used once.

	8	9	10
Cy			
Di			
Ed			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q4
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three people and three snacks, one each. Gil does not have the apple or the pear. Which snack does Gil have, if the snacks are apple, pear and plum?

A. apple
B. it cannot be known
C. pear
D. plum

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q5
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four friends, Ari, Bea, Cal and Dev, finished a quiz with scores 6, 7, 8 and 9 (all different). Bea scored 1 more than Dev. Ari scored the most. Cal did not score 6. What did each score?

Choose one column for each row. Each column is used once.

	6	7	8	9
Ari				
Bea				
Cal				
Dev				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q1
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Five boxes in a row, positions 1 to 5, hold a hat, a kite, a lamp, a mug and a net (one each). The kite is in an even position. The lamp is directly left of the mug. The hat is at position 5. The net is not next to the kite. Which item is in each position?

Choose one column for each row. Each column is used once.

	hat	kite	lamp	mug	net
1					
2					
3					
4					
5					

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q2
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Rae, Sol and Tom own a cat, a dog and a fish, one each. Clue 1: Rae does not own the cat. Clue 2: Sol does not own the dog. These two clues allow three matchings. Which additional clue leaves exactly one?

A. Rae does not own the fish.

B. Tom owns the cat.

C. Tom does not own the dog.

D. Sol does not own the fish.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q3
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three students, Uma, Vik and Wyn, each chose a different sport (golf, judo or swimming) and a different day (Monday, Tuesday, Wednesday). The judo student trains on Monday. Uma trains on Wednesday. Uma does not play golf. Wyn trains on Tuesday. Which sport did each choose?

Choose one column for each row. Each column is used once.

	golf	judo	swimming
Uma			
Vik			
Wyn			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q4
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a matching puzzle with 4 people and 4 options (one each), if three people's options are known, the fourth person's option is also known.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q5
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four students, Al, Bo, Cy and Di, each have a different favorite number from 1 to 4 and a different color (red, blue, green, gold). The red student's number is 1 more than the blue student's number. Al is gold. Bo's number is 4. Cy's number is twice Di's number. What is each student's number?

Choose one column for each row. Each column is used once.

	1	2	3	4
Al				
Bo				
Cy				
Di				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q1
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three people wear a red, a blue and a green hat, one each, so there are 6 possible matchings. Clues have the form "person X does not wear color Y". What is the smallest number of such clues that can leave exactly one matching?

A. 3
B. 4
C. 1
D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q2
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four people each get a different one of 4 prizes. How many matchings are possible if one person, Ash, cannot get prize 1 or prize 2?

Matchings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q3
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a 3-person, 3-option matching puzzle, the clues "A not x" and "B not x" together force C to have x.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q4
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Five players wear the numbers 1 to 5 (all different). Al's number plus Bo's number is 9. Give one possible pair: Al's number a and Bo's number b.

Al a = _____

Bo b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q5
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Set 15: Converse, inverse and contrapositive

Local outcome LOGIC.CONDITIONAL.03: Form the converse, inverse and contrapositive and decide which are equivalent to the original

Formal logic set · genre: formal logic · MP3, MP7

MAIN: Form the converse, inverse and contrapositive and judge equivalence
BACKWARD 1: Negate each part
BACKWARD 2: Hypothesis and conclusion
FORWARD 1: ADVANCED APPLICATION: Which forms are equivalent?
FORWARD 2: EXPERT CHALLENGE: Converses that hold and fail

What is the CONTRAPOSITIVE of "If a shape is a triangle, then it has three sides"?

A. A shape is a triangle and does not have three sides.
B. If a shape is not a triangle, then it does not have three sides.
C. If a shape does not have three sides, then it is not a triangle.
D. If a shape has three sides, then it is a triangle.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q1
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

Which statement is logically EQUIVALENT to "If n is a multiple of 4, then n is even"?

A. If n is even, then n is a multiple of 4.
B. n is a multiple of 4 and n is even.
C. If n is not a multiple of 4, then n is not even.
D. If n is not even, then n is a multiple of 4.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q2
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

The converse of "If a whole number is a multiple of 6, then it is even" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q3
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

The statement "If x > 2, then x squared > 4" is true. Find a real number x that shows its CONVERSE is false.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q4
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

"If P, then Q" is true. Which statement MUST also be true?

A. If not Q, then not P.
B. P and Q.
C. If Q, then P.
D. If not P, then not Q.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q5
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

What is the negation of "the shape has three sides"?

A. the shape has no sides
B. the shape has at least three sides
C. the shape does not have three sides
D. the shape has four sides

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q1
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

What is the negation of "n is even" for a whole number n?

A. n is zero
B. n is odd
C. n is prime
D. n is negative

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q2
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

The negation of "x > 3" is "x < 3".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q3
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

What is the negation of "it is raining"?

A. it is not raining
B. it will rain
C. it is sunny
D. it rained yesterday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q4
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

"Not (not P)" says the same as "not P".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q5
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

In "If it snows, school is closed," what is the conclusion?

A. school is open
B. school is closed
C. it snows
D. it is cold

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q1
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Swap the parts of "If P, then Q". What do you get?

A. If not Q, then not P.

B. P and Q.

C. If not P, then not Q.

D. If Q, then P.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q2
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

In "If x = 5, then x + 1 = 6," the conclusion is "x = 5".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q3
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Which statement has hypothesis "a figure is a square"?

A. Every rectangle is a square.
B. A figure is a rectangle only if it has four sides.
C. If a figure is a square, then it is a rectangle.
D. If a figure is a rectangle, then it is a square.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q4
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Swapping and negating both parts of "If P, then Q" gives "If not Q, then not P".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q5
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Complete the truth table for "if P then Q" and its contrapositive "if not Q then not P".

Fill in T or F in every empty cell.

P	Q	if P then Q	if not Q then not P
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q1
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

Which two statements are ALWAYS equivalent to each other?

A. the converse and the inverse
B. the original and the converse
C. the original and the inverse
D. the converse and the contrapositive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q2
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

For how many whole numbers n from 1 to 20 is the INVERSE of "If n is a multiple of 4, then n is even" false?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q3
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

If a conditional is false, its contrapositive is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q4
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

"If a quadrilateral is a square, then its diagonals are equal." Which related statement is FALSE?

A. the contrapositive: if the diagonals are not equal, it is not a square

B. the converse: if the diagonals are equal, the quadrilateral is a square

C. the original statement

D. "a square has equal diagonals"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q5
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

For whole numbers, both "If n is odd, then n squared is odd" and its converse are true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q1
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

The statement "If n is prime and greater than 2, then n is odd" is true. Its converse is "If n is odd, then n is prime and greater than 2". Give a whole number n that shows the converse is false.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q2
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

A statement and its converse are both true. Which statement must be true?

A. P or Q

B. not P

C. P if and only if Q

D. P and Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q3
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

In how many of the 4 rows of a truth table do "if P then Q" and its converse "if Q then P" have DIFFERENT truth values?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q4
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

C is the conditional "if P then Q". Which statement is correct?

A. If C is false, its converse is false.

B. If C is true, its contrapositive may be false.

C. If C is true, its converse is true.

D. If C is true, its inverse may be true or false.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q5
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

Set 16: Biconditionals: "if and only if"

Local outcome LOGIC.CONDITIONAL.04: Find truth values of biconditionals and use them as definitions

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Find truth values of "if and only if" statements and use them as definitions
BACKWARD 1: Conditionals first
BACKWARD 2: The converse
FORWARD 1: ADVANCED APPLICATION: Biconditionals as definitions and tests
FORWARD 2: EXPERT CHALLENGE: Chains and opposites of biconditionals

Complete the truth table for "P if and only if Q".

Fill in T or F in every empty cell.

P	Q	P if and only if Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q1
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

For real numbers x, "x squared = 9 if and only if x = 3" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q2
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Find a real number x that makes "x > 5 if and only if x > 3" FALSE.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q3
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Which statement is a correct DEFINITION of an even whole number?

A. A whole number is even only if it is greater than 1.
B. A whole number is even if and only if it is divisible by 2.
C. A whole number is even if and only if it ends in 0.
D. A whole number is even if it is divisible by 4.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q4
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"P if and only if Q" is true and P is false. What follows?

A. Nothing about Q.
B. Q is true.
C. Q is false.
D. P and Q are both true.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q5
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"If 2 > 3, then 4 > 5" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q1
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"If 3 > 2, then 4 > 5" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q2
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

When is "if P then Q" false?

A. when P is true and Q is false
B. when both are false
C. when Q is true
D. when P is false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q3
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"If 4 > 1, then 2 > 6" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q4
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

In how many of the 4 rows is "if P then Q" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q5
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

What is the converse of "If it is a dog, then it barks"?

A. It is a barking dog.
B. If it is not a dog, then it does not bark.
C. If it does not bark, then it is not a dog.
D. If it barks, then it is a dog.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q1
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

The converse of "If n = 2, then n is even" is "If n is not 2, then n is not even".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q2
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

The converse of a true statement is always true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q3
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

"P if and only if Q" means "if P then Q" AND what?

A. if Q then P
B. P or Q
C. if not P then Q
D. not P and not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q4
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

If a conditional is true but its converse is false, the biconditional is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q5
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

For how many whole numbers n from 1 to 20 is "n is even if and only if n is a multiple of 4" TRUE?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q1
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Complete the table to compare "P if and only if Q" with "(if P then Q) and (if Q then P)".

Fill in T or F in every empty cell.

P	Q	P if and only if Q	(if P then Q) and (if Q then P)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q2
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Which definition of a SQUARE is correct (works in both directions)?

A. A quadrilateral is a square if and only if its diagonals are equal.
B. A quadrilateral is a square if and only if it has four equal sides and four right angles.
C. A quadrilateral is a square if and only if it has four right angles.
D. A quadrilateral is a square if and only if it has four equal sides.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q3
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"A whole number is divisible by 10 if and only if it ends in 0" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q4
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Find a whole number n that makes "n is divisible by 3 if and only if n is odd" FALSE.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q5
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"P exclusive-or Q" has the opposite truth value of "P if and only if Q" in every row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q1
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"P iff Q" and "Q iff R" are both true. Which must be true?

A. P exclusive-or R
B. P and R
C. P iff R
D. not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q2
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

In how many of the 8 rows of a truth table for P, Q, R is "(P iff Q) iff R" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q3
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Find whole numbers a and b, each from 1 to 9, that make "a + b is even if and only if a is even" TRUE and a different from b.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q4
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Which statement about whole numbers is TRUE?

A. n is even if and only if n is a multiple of 4.
B. n is a multiple of 3 if and only if n is a multiple of 9.
C. n is odd if and only if n is prime.
D. n is odd if and only if n + 1 is even.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q5
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Set 17: Truth-tellers and liars

Local outcome LOGIC.KNIGHTS.01: Find who tells the truth when each speaker either always tells the truth or always lies

Puzzle set · genre: truth-teller puzzles · MP1, MP3

MAIN: Decide who tells the truth when each speaker always tells the truth or always lies
BACKWARD 1: Strategy: suppose, then test
FORWARD 1: ADVANCED APPLICATION: Statements about counts
FORWARD 2: EXPERT CHALLENGE: Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ana says: "Ben is a liar." Ben says: "Ana and I are both truth-tellers." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Ana		
Ben		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q1
Logic, Set 17: Truth-tellers and liars · MAIN · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Cal says: "Dee and I are different kinds." Dee says: "Cal is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Cal		
Dee		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q2
Logic, Set 17: Truth-tellers and liars · MAIN · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Eve says: "Fin and Gia are both liars." Fin says: "Eve is a truth-teller." Gia says: "Fin is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Eve		
Fin		
Gia		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q3
Logic, Set 17: Truth-tellers and liars · MAIN · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Jo says: "Kai is a liar." Kai says: "Lu is a liar." Lu says: "Jo and Kai are both liars." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Jo		
Kai		
Lu		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q4
Logic, Set 17: Truth-tellers and liars · MAIN · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Mo says: "Exactly one of Ned and Ola is a liar." Ned says: "Mo is a liar." Ola says: "Ned and I are the same kind." Who is a liar?

A. Ned only
B. Mo only
C. Mo and Ned
D. Ola only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q5
Logic, Set 17: Truth-tellers and liars · MAIN · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A truth-teller can say "I am a liar."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q1
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A liar can say "I am a truth-teller."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q2
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Vic says: "2 + 2 = 5." What is Vic?

A. either
B. neither
C. a liar
D. a truth-teller

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q3
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Wes is a truth-teller and says: "Xia is a liar." What is Xia?

A. neither
B. a truth-teller
C. either
D. a liar

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q4
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Yan is a liar and says: "Zed is a liar." What is Zed?

A. a truth-teller
B. a liar
C. either
D. neither

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q5
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ash says: "Exactly one of us three is a truth-teller." Bly says: "Exactly two of us three are truth-tellers." Cru says: "None of us three is a truth-teller." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Ash		
Bly		
Cru		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q1
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Dov says: "At least one of Ela and Fyn is a liar." Ela says: "Dov and Fyn are both truth-tellers." Fyn says: "Dov is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Dov		
Ela		
Fyn		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q2
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Pat says: "Exactly two of us three are truth-tellers." Quy says: "Pat is a liar." Ren says: "Quy and Pat are the same kind." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Pat		
Quy		
Ren		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q3
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. If two people each say "The other one is a liar," then exactly one of them is a liar.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q4
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Four people each say exactly the same sentence: "Exactly one of us four is a truth-teller." How many of them are truth-tellers?

Truth-tellers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q5
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Uri says: "Exactly three of us four are liars." Vee says: "Uri is a truth-teller." Wil says: "Vee is a liar." Xan says: "Wil is a truth-teller and Uri is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Uri		
Vee		
Wil		
Xan		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q1
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Which statement can NO person on the island say?

A. "I am a liar."
B. "2 + 2 = 5."
C. "2 + 2 = 4."
D. "I am a truth-teller."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q2
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Five people stand in a circle. Each says: "The person on my left is a liar." In how many ways can the five be truth-tellers and liars so that every statement fits?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q3
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A person says: "I am a liar or 2 + 2 = 4." This person must be a truth-teller.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q4
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. You meet Gil and Hal. Gil says: "At least one of us two is a truth-teller." Hal says nothing. Which statement MUST be true?

A. Hal is a liar.
B. Gil is a truth-teller.
C. Both are liars.
D. If Hal is a truth-teller, then Gil is a truth-teller.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q5
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

Set 18: Truth tables

Local outcome LOGIC.TRUTH.01: Complete truth tables for compound statements

Formal logic set · genre: formal logic · MP6, MP8

- MAIN: Complete truth tables for compound statements
- BACKWARD 1: The basic connectives
- BACKWARD 2: Work from the inside out
- FORWARD 1: ADVANCED APPLICATION: Three-variable tables
- FORWARD 2: EXPERT CHALLENGE: Read and build statements from tables

Complete the truth table for "P and (not Q)".

Fill in T or F in every empty cell.

P	Q	not Q	P and (not Q)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q1
Logic, Set 18: Truth tables · MAIN · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(not P) or Q".

Fill in T or F in every empty cell.

P	Q	not P	(not P) or Q
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q2
Logic, Set 18: Truth tables · MAIN · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P or Q) and not (P and Q)".

Fill in T or F in every empty cell.

P	Q	P or Q	P and Q	(P or Q) and not (P and Q)
T	T			
T	F			
F	T			
F	F			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q3
Logic, Set 18: Truth tables · MAIN · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A truth table for three statements P, Q and R lists every combination of truth values. How many rows does it have?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q4
Logic, Set 18: Truth tables · MAIN · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P and Q) or R".

Fill in T or F in every empty cell.

P	Q	R	P and Q	(P and Q) or R
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q5
Logic, Set 18: Truth tables · MAIN · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "P and Q".

Fill in T or F in every empty cell.

P	Q	P and Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q1
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

Complete the truth table for "P or Q" (inclusive).

Fill in T or F in every empty cell.

P	Q	P or Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q2
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

Complete the truth table for "not P".

Fill in T or F in every empty cell.

P	not P
T	
F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q3
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

"P and Q" is true when P is true and Q is false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q4
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

"P or Q" is true when P is false and Q is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q5
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

In "not (P and Q)", which part is computed first?

A. not P
B. not Q
C. P and Q
D. P or Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q1
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

"not (P and Q)" and "(not P) and Q" always have the same truth value.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q2
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

P is true and Q is false. Is "not (P and Q)" true or false? Enter 1 for true or 0 for false.

Value (1 or 0): _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q3
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

P is false, Q is true. What is the value of "(P or Q) and P"?

A. true
B. false
C. both
D. it cannot be found

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q4
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "not (P or Q)".

Fill in T or F in every empty cell.

P	Q	P or Q	not (P or Q)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q5
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "P and (Q or R)".

Fill in T or F in every empty cell.

P	Q	R	Q or R	P and (Q or R)
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q1
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P or Q) and (not R)".

Fill in T or F in every empty cell.

P	Q	R	P or Q	(P or Q) and (not R)
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q2
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

In how many of the 8 rows is "P or (Q and R)" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q3
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A statement in P and Q is true only in the row P true, Q false. Which statement is it?

A. P and (not Q)
B. not (P and Q)
C. P or (not Q)
D. (not P) and Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q4
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A truth table for 4 statements has 16 rows.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q5
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A statement in P, Q, R is true in exactly the rows where an odd number of P, Q, R are true. Which statement is it?

A. P or Q or R
B. P and Q and R
C. P exclusive-or (Q exclusive-or R)
D. (P and Q) or R

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q1
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

How many different possible output columns can a truth table with two statement letters P and Q have (4 rows, each output T or F)?

Columns: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q2
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P and not Q) or (not P and Q)".

Fill in T or F in every empty cell.

P	Q	P and not Q	not P and Q	(P and not Q) or (not P and Q)
T	T			
T	F			
F	T			
F	F			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q3
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

In how many of the 16 rows of a table for P, Q, R, S is "(P and Q) or (R and S)" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q4
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Every truth-table column for P and Q can be produced by some statement using only "not", "and", "or".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q5
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8